

Chapter 10

Dhandho 301: Few Bets, Big Bets, Infrequent Bets

Let's assume you were offered the following odds on a \$1 bet:

80 percent chance of winning \$21.00

10 percent chance of winning \$7.50

10 percent chance of losing it all

Let's further assume that you had \$10,000 to your name and you were allowed to bet as much of that bankroll as you wanted. How much of that \$10,000 would you be willing to put at stake to play this game once? The answer is clearly not \$10,000, as there is a solid 10 percent chance of being in the poorhouse. Betting \$1 seems too conservative—it isn't going to move the needle.

The good news is that exactly 50 years ago a researcher at Bell Labs in New Jersey, Mr. John Larry Kelly Jr. pondered this question and published his findings. Kelly came up with what is now known as the Kelly Formula. Kelly calculated that the optimal fraction of your bankroll to bet on a favorable bet is:

$$\text{Edge/odds} = \text{Fraction of your bankroll you should bet each time}$$

There is a wonderful book written by William Poundstone entitled *Fortune's Formula*¹ that is well worth reading. Poundstone describes the Kelly Formula beautifully. Michael Mauboussin of Legg Mason recently wrote a paper² on the Kelly Formula where he used the following illustration: assume you're offered a coin toss where heads means you get